

Diffygons

Fall 2019 ARML Power Contest

The notation used throughout this set of problems will be as it was in the background handout: If $A = (a_1, a_2, \dots, a_{n-1}, a_n)$ is an ordered n -tuple, then

$$\Delta A = (|a_1 - a_2|, |a_2 - a_3|, \dots, |a_{n-1} - a_n|, |a_n - a_1|)$$

is its differences, and repeated applications of Δ are indicated by superscripts. Please make sure to keep lists of differences in their proper order! A diffy *box* will always be a diffygon with exactly four sides (and we will, of course, just use ordered lists of four numbers); diffy*gons* can be lists of any length n , representing polygons with numbers at each of their n vertices.

Problem 1. The diffy box $A = (9, 17, 31, 57)$ has one of the largest longevities among all diffy boxes with four positive integers less than 100.

- [1] (a) Compute $\Delta^4 A$.
- [1] (b) Compute the longevity of A .

Given that $A = (a_1, a_2, \dots, a_n)$ is an ordered n -tuple, and that c and k are real numbers, the notation $A + c$ will mean to add c to each term A and $k \cdot A$ will mean to multiply each term of A by k . That is, $A + c = (a_1 + c, a_2 + c, \dots, a_n + c)$ and $k \cdot A = (ka_1, ka_2, \dots, ka_n)$.

Problem 2. Let $A = (4, 2, 8, 3, 6)$.

- [1] (a) Compute ΔA .
- [1] (b) Compute $3 \cdot A + 7$.
- [1] (c) Compute $\Delta(3 \cdot A + 7)$.

You might have noticed that the result of part (c) of the previous problem is simply 3 times the result of part (a). This is no accident.

Problem 3. Let $A = (a_1, a_2, \dots, a_n)$ be an arbitrary n -tuple of real numbers.

- [2] (a) Prove that $\Delta(A + c) = \Delta A$ for any real number c .
- [2] (b) Prove that $\Delta(k \cdot A) = |k| \cdot \Delta A$ for any real number k .

[2] **Problem 4.** Let $A = (a_1, a_2, \dots, a_n)$ be an arbitrary n -tuple of real numbers. Show that there is an n -tuple B whose minimum element is 0 and for which $\Delta B = \Delta A$.

If A is an n -tuple, define its *range*, notated $r(A)$ to be the difference between its maximum and minimum elements, whether or not that difference actually appears in its list of differences. Thus, the range of $(2, 6, 0, -3, 4)$ is $r(2, 6, 0, -3, 4) = 6 - (-3) = 9$.

Problem 5.

- [2] (a) Prove that $r(A + c) = r(A)$.
- [2] (b) Prove that $r(k \cdot A) = |k| \cdot r(A)$.
- [2] (c) Prove that $r(\Delta A) \leq r(A)$.

Amazingly, even with arbitrary, seemingly unrelated real numbers as their entries, diffy boxes tend to end up at all zeroes after a short amount of time.

- [2] **Problem 6.** Begin with $A = (\pi, e, 4, \sqrt{2})$. Compute the results of repeatedly applying Δ to A until it becomes all zeroes. (Make sure to show the results of each step; if you don't reach all zeroes after a few applications of Δ you have made a computational error.)

If you have played around with diffy boxes, you may have noticed that they *always* seem to go to zero eventually. This is *almost* true, even (as you just saw) if the entries aren't integers! This doesn't happen for all numbers of sides of diffygons (it happens when the number of sides is a power of two, to be precise) and the proof when the entries are not all integers is difficult. But for integer entries and diffy boxes with four sides you will only need one more piece:

- [2] **Problem 7.** Given that $A = (a_1, a_2, a_3, a_4)$ is a list of four integers, show that all entries of $\Delta^4 A$ are even integers.

- [4] **Problem 8.** Prove that the longevity of any diffy box of four integers is finite.

Define the *Tribonacci* numbers as follows: $t_0 = 0, t_1 = 0, t_2 = 1$, and for $n > 2, t_n = t_{n-1} + t_{n-2} + t_{n-3}$. The sequence is similar to the Fibonacci sequence except that *three* consecutive numbers are added instead of just two. Thus the sequence starts $0, 0, 1, 1, 2, 4, 7, 13, 24, 44, \dots$. Let T_n be the diffy box $(t_n, t_{n+1}, t_{n+2}, t_{n+3})$.

- [2] **Problem 9.** Given that $n \geq 2$, prove that $\Delta^3 T_n = 2 \cdot T_{n-2}$.

- [2] **Problem 10.** Compute an exact formula of n for the longevity of T_n .

The result of the last problem shows that there are diffy boxes that can last arbitrarily long before becoming all zeroes. Allowing *real* entries, there are diffy boxes that can last forever.

- [2] **Problem 11.** Let t be the unique real root of the cubic polynomial $t^3 - t^2 - t - 1 = 0$. (The value of t is approximately 1.839.) Show that $\Delta(1, t, t^2, t^3) = (t - 1) \cdot (1, t, t^2, t^3)$.

Thus, the diffy box $(1, t, t^2, t^3)$ lasts forever, with every application of Δ simply multiplying the previous result by $t - 1$. Of course, since $0 < t - 1 < 1$, repeatedly multiplying by $t - 1$ causes the result to have a limit of zero, even though it never actually reaches a sequence of all zeroes. It can be shown that a diffy box of four real numbers has a finite longevity if and only if it is not a rotation or reflection of $k \cdot (1, t, t^2, t^3) + c$ for some real numbers k and c .

The situation is very different for diffy triangles and other shapes!

Problem 12. Let $A = (a_1, a_2, \dots, a_n)$ be an ordered n -tuple of real numbers representing a diffygon with n sides.

- [1] (a) Show that the longevity of A is 1 if and only if A is constant and not already all zeroes (that is, $a_1 = a_2 = \dots = a_n \neq 0$).
- [3] (b) Show that if n is odd then there are *no* diffygons with n entries with finite longevity ≥ 2 .

For n -sided diffygons, if n is a power of 2 then the behavior seen for diffy boxes ($n = 4 = 2^2$) dominates. That is, almost all such diffygons have finite longevity. The proof for integer diffygons is basically identical to the proof contained in this Power Contest, although it takes more steps to guarantee all even terms. For arbitrary real numbers, almost all diffygons have finite longevity, with exceptions similar to the one demonstrated above for $n = 4$.

When n has an odd factor, however, the behavior of the previous problem takes over, and unless the diffygon can be split into identical chunks that are a power of 2 in length (for example, $(1, 4, 6, 3, 1, 4, 6, 3, 1, 4, 6, 3)$ which repeats the chunk $(1, 4, 6, 3)$ of length 4), its repeated differences will fall into a repeating pattern (as happened in the final question of the background sheet).

Problems 10 and 11 showed that there are diffy boxes that have arbitrarily large or even infinite longevity. This can only happen if there is one corner of the box and a direction to go around the box starting in that corner for which the entries appear in increasing order. Otherwise, the longevity can never be more than six!

- [5] **Problem 13.** Prove this last statement, by classifying all the non-increasing diffy boxes according to their longevity. That is, we have already shown that a diffy box has longevity 1 if and only if its entries are all the same non-zero constant. It is easy to see that (a, b, a, b) has longevity 2 for any real a and b ; what other boxes will have longevity 2? Longevity 3? By problem 4 you can shift the box so that its minimum entry is zero, if convenient. By the end of your classification, you should have included all boxes $(0, a, b, c)$ that do not satisfy $0 < a < b < c$ and shown that all such boxes have longevity six or less.